NUKE: The Automotive Data API

Strategic Positioning & Product Report

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Prepared: February 2026 **Classification:** Internal — Founder's Eyes Only **Version:** 1.0

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1. Executive Summary

Nuke is positioning as **the definitive API for automotive data** — not an AI company that happens to have cars, but a data company whose product is the most complete, most traceable, most trustworthy automotive dataset ever assembled.

The automotive data market is estimated at $6-8B globally and growing at 15-20% CAGR, driven by fintech, insurtech, dealer SaaS, and the explosion of online vehicle marketplaces. Yet the existing providers — KBB, Black Book, J.D. Power, Edmunds — are built on decades-old models: editorial estimates, dealer surveys, guidebook valuations. They tell you what a vehicle *should* be worth. They don't tell you what vehicles *actually* sold for, where, when, and with what level of confidence.

Nuke does.

Current state as of February 2026: - 812,913 vehicles in the database - 30.4 million images - 507,071 computed valuations (62.4% coverage) - 617,385 observations from 517 active sources - 132,630 indexed BaT listings with 11M+ auction comments - 22,775 new vehicles ingested in the last 24 hours - 25+ platform integrations across auctions, classifieds, and marketplaces - Published OpenAPI 3.0 spec, TypeScript SDK, webhook system

The infrastructure is built. The data is flowing. The question now is packaging and positioning.

2. The Thesis

Every other automotive data provider is an opinion. Nuke is a record.

Traditional automotive valuation works like this: an editorial team surveys dealers, reviews wholesale auction data (Manheim, ADESA), applies adjustment factors, and publishes a “value.” This number is an estimate — a blend of methodology, judgment, and update frequency (often monthly or quarterly).

Nuke works differently. Every piece of data in the system is an **observation** — a discrete, timestamped, source-attributed data point. A BaT hammer price is an observation. A Craigslist asking price is an observation. A Facebook Marketplace listing that sat for 47 days and sold for 15% below ask is an observation. A forum post where someone mentions buying their car for $12,000 in 2019 is an observation.

These observations are: - **Immutable** — once recorded, they cannot be changed - **Source-attributed** — every observation carries its origin (platform, URL, extraction method) - **Confidence-scored** — every observation has a trust rating based on source reliability and corroboration - **Time-ordered** — creating a true history of the vehicle, not a snapshot

This is not a better opinion. It is a different category of product.

The thesis is simple: in a world of increasingly sophisticated data consumers — algorithmic lenders, programmatic insurance underwriters, AI-powered dealer tools — the winner is whoever can provide the rawest, most traceable, most comprehensive automotive transaction data. Not summaries. Not estimates. Observations.

3. Market Landscape

3.1 Market Size

The global automotive data market spans several overlapping segments:

Segment	Estimated Size	Growth Rate
Vehicle valuation & pricing data	$2.5B	12-15%
VIN decoding & specification data	$800M	8-10%
Vehicle history reports	$1.5B	10-12%
Dealer analytics & market intelligence	$1.2B	18-22%
Insurance & lending vehicle data	$1.8B	15-18%
Total addressable	**~$7.8B**	**~15%**

The fastest-growing segments are dealer analytics (driven by margin compression and inventory optimization needs) and insurance/lending data (driven by algorithmic underwriting and embedded finance).

3.2 Market Structure

The market is currently structured around **use-case-specific providers** rather than general-purpose data APIs:

- **Valuation**: KBB, Black Book, J.D. Power (NADA), Edmunds
- **VIN/Specs**: DataOne, Chrome Data (Autodata), NHTSA
- **History**: CarFax, AutoCheck (Experian)
- **Listings/Inventory**: MarketCheck, AutoDev, Cars.com
- **Enthusiast/Classic**: Hagerty, Classic.com, BaT (raw data)

No single provider offers a unified, source-attributed dataset spanning all of these. Every consumer stitches together 3-5 providers, pays for redundant coverage, and reconciles conflicting data manually.

3.3 Market Trends

1. Programmatic consumption is replacing manual lookup. Dealers used to open KBB in a browser. Now their DMS calls an API. Lenders used to pull a NADA guide. Now their decisioning engine calls a valuation API in the loan origination flow.

2. Classic and enthusiast vehicles are underserved. KBB and Black Book focus on vehicles 1-20 years old. The collector market ($30B+ in annual transactions) has no equivalent data infrastructure. Hagerty's price guide is editorial. BaT results are public but unstructured.

3. Data provenance is becoming a requirement. Regulators, auditors, and institutional buyers increasingly require explainability — not just "the car is worth $X" but "here's why, based on these transactions, with this confidence level."

4. Real-time is replacing periodic. Monthly guidebook updates are too slow for a market where auction results, listing prices, and market sentiment change daily.

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4. Competitive Analysis

4.1 Kelley Blue Book (KBB) / Cox Automotive

What they offer: Consumer-facing and API-based vehicle valuations for vehicles 1-20 years old. Trade-in, private party, and dealer retail values. Powers many dealer and lending integrations.

Strengths: Brand recognition (96% consumer awareness). Massive dealer network through Cox (Autotrader, Dealer.com, vAuto). Integration with major DMS platforms.

Weaknesses: Valuation methodology is opaque — editorial + algorithmic blend with no source attribution. Poor coverage of vehicles >20 years old. No auction transaction data. API access is enterprise-only with long sales cycles. No real-time data. Values update periodically, not continuously.

Pricing: Enterprise contracts, typically $50K-$500K/year depending on volume and use case. No self-serve API.

Gap Nuke fills: Source-level transparency, real-time auction data, classic/enthusiast coverage, self-serve API access.

4.2 Black Book

What they offer: Wholesale and retail vehicle valuations, primarily serving dealers, lenders, and insurers. Strong in wholesale (Manheim/ADESA auction data). Daily value updates.

Strengths: Deep wholesale auction integration. Daily updates (faster than KBB). Strong in lending — used by many auto lenders for loan-to-value calculations. Condition-adjusted values.

Weaknesses: Wholesale-focused — less relevant for retail/consumer transactions. No enthusiast/classic coverage. No retail auction data (BaT, C&B, etc.). Opaque methodology. Enterprise pricing only.

Pricing: Enterprise contracts, $30K-$200K/year. Per-VIN pricing available for some tiers ($0.10-$0.50/lookup).

Gap Nuke fills: Retail auction transparency, enthusiast/classic data, observation-level detail, accessible pricing.

4.3 J.D. Power (formerly NADA Guides)

What they offer: Vehicle valuation guides used extensively in lending and insurance. The “NADA value” is the standard for many credit unions and insurance companies.

Strengths: Regulatory acceptance — many lenders are contractually required to use NADA values. Long history and institutional trust. Good condition adjustment methodology.

Weaknesses: Extremely slow update cycle. Conservative valuations (by design — lenders want downside protection). No transaction-level data. No auction data. Classic coverage is minimal. API is dated.

Pricing: Enterprise contracts, often bundled with J.D. Power analytics.

Gap Nuke fills: Transaction-level data for audit trails, real-time values, enthusiast/classic coverage.

4.4 Edmunds

What they offer: Consumer-facing vehicle information, reviews, and pricing. TMV (True Market Value) pricing. API program for inventory and pricing data.

Strengths: Strong consumer brand. Good editorial content. Decent API program with self-serve options. New car transaction data from dealer partners.

Weaknesses: Primarily new/near-new vehicles. No auction data. No classic/enthusiast coverage. TMV is modeled, not observed. API has been deprioritized (limited updates in recent years).

Pricing: API plans from $0 (limited) to custom enterprise.

Gap Nuke fills: Used/classic vehicle depth, auction data, observation model.

4.5 MarketCheck

What they offer: Automotive listing aggregation API. Real-time inventory data from dealer websites, classifieds, and marketplaces. VIN decoding. Pricing analytics.

Strengths: Good listing coverage (~5M active listings claimed). Clean API design. Self-serve access. Reasonable pricing. Active development.

Weaknesses: Listing data only — no auction results, no transaction history, no sold prices. No provenance tracking. No classic/enthusiast focus. Aggregation without confidence scoring.

Pricing: From $99/mo (5K requests) to $999/mo (100K requests). Enterprise custom.

Gap Nuke fills: Sold prices, auction data, confidence scoring, provenance, classic coverage.

4.6 Hagerty

What they offer: Hagerty Valuation Tools (HVT) — the de facto pricing guide for collector vehicles. Insurance-focused. Some API access for partners. Market trend reports.

Strengths: Best brand in classic/collector segment. Extensive editorial expertise. Good condition-adjusted valuations for classics. Large insured vehicle dataset.

Weaknesses: Editorial valuations (expert opinion, not transaction data). Slow update cycles. Limited API access (partner-only). No real-time auction data despite owning auction platforms. No cross-platform aggregation. No source attribution — just a number.

Pricing: Partner API access by agreement. Consumer access via website/app.

Gap Nuke fills: Transaction-based valuations instead of editorial, cross-platform auction aggregation, source attribution, real-time updates.

4.7 Classic.com

What they offer: Auction result aggregation for classic and collector vehicles. Price history charts. Market analytics.

Strengths: Good historical auction result coverage. Clean UI. Growing dataset. Some API access.

Weaknesses: Aggregation only — limited original data extraction. No observation model. No confidence scoring. No classified/marketplace data. API is limited.

Pricing: Subscription-based consumer product. API pricing not publicly documented.

Gap Nuke fills: Broader source coverage (classifieds, marketplaces, not just auctions), observation model, confidence scoring, richer data per vehicle.

4.8 Drivly

What they offer: "The Stripe for Car Buying" — APIs for VIN decode, vehicle data, pricing, and automotive commerce workflows.

Strengths: Developer-friendly positioning. Modern API design. Multi-endpoint approach (similar to what Nuke should do). Good marketing.

Weaknesses: Thin data layer — largely aggregating from other providers (NHTSA, etc.) rather than original sourcing. No auction data. No observation model. No classic coverage. Limited proven scale.

Pricing: Self-serve, usage-based. Approx $0.05-$0.15 per API call depending on endpoint.

Gap Nuke fills: Original source data instead of re-aggregation, auction depth, classic coverage, provenance.

4.9 CarFax / AutoCheck

What they offer: Vehicle history reports — accident history, title events, odometer readings, service records.

Strengths: Massive brand (CarFax). Deep DMV/insurance/service data partnerships. Near-universal adoption in dealer transactions.

Weaknesses: History-focused, not valuation-focused. No pricing data. No auction data. No market analytics. API access is expensive and restricted. Per-report pricing ($40+ consumer).

Gap Nuke fills: Different product category — Nuke is complementary, not competitive. Could integrate CarFax data as another observation source.

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5. Where Everyone Else Falls Short

Every incumbent automotive data provider shares the same structural weakness: **they don't show their work.**

When KBB says a 2015 Ford F-150 is worth $28,500, you get a number. You don't get: - Which transactions informed that number - How recent those transactions are - Whether the comparable vehicles were actually comparable (same trim, same condition, same region) - How confident the provider is in the estimate - What the spread of actual transaction prices looks like

When Hagerty says a 1970 Chevelle SS 396 in Condition 2 is worth $78,000, you get an editorial opinion. You don't get: - The actual auction results that inform it - The asking prices on current marketplace listings - The sentiment from auction comment sections - Whether that number was updated last week or last quarter - How many actual sales that estimate is based on

This opacity was acceptable when the consumer was a human making a judgment call. A dealer glancing at KBB to set a trade-in value can apply their own experience and intuition.

It is unacceptable when the consumer is an algorithm. A lending model, an insurance underwriting engine, a dealer pricing optimizer, an investment fund — these consumers need:

1. **Raw transaction data** to feed their own models
2. **Source attribution** for regulatory compliance and audit
3. **Confidence intervals** to calibrate risk
4. **Real-time updates** to avoid stale pricing
5. **Coverage of the full market** — not just new/near-new or just classics

No existing provider delivers all five. Nuke does.

6. What Nuke Has Built

6.1 The Platform

Nuke is a full-stack automotive data platform consisting of:

Component	Technology	Purpose
Data Pipeline	312 Supabase Edge Functions (Deno/TypeScript)	Extraction, processing, enrichment
Database	PostgreSQL via Supabase	687 migrations, RLS-secured
Backend API	Elixir/Phoenix	RESTful endpoints, rate limiting
Frontend	React + Vite	Consumer interface (n-zero.dev)
SDK	TypeScript (@nuke/sdk)	Developer client library
Mobile	Nuke Scanner, Nuke Desktop	Field data collection
Sub-product	DealerScan	Dealer document OCR
AI Layer	Claude, GPT, Gemini, Ollama	Extraction, analysis, sentiment
MCP Server	Model Context Protocol	AI-native data access

6.2 What Makes It Different from a Scraper

Many startups have built automotive scrapers. Nuke is not a scraper — it is a **data platform with an observation-based architecture**. The distinction matters:

A scraper collects listing data, stores it in a table, and serves it. When the listing changes, the old data is overwritten. There is no history, no provenance, no confidence model.

Nuke's observation model treats every piece of information as a discrete event: - An observation is created when data is extracted from a source - The observation is immutable — it is never modified after creation - Multiple observations about the same vehicle create a timeline - Conflicting observations are not resolved by overwriting — they coexist with different confidence scores - The consumer (human or algorithm) can see all observations and decide which to trust

This is the difference between a database and a ledger. Nuke is a ledger.

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7. The Data Advantage

7.1 Breadth of Sources

Nuke ingests data from 25+ platforms across four categories:

Auction Houses (15 platforms)

Platform	Coverage	Data Depth
Bring a Trailer	132,630 listings, 11M+ comments	Full: bids, comments, sentiment, images
Cars & Bids	Active extraction	Listings, bids, comments
RM Sotheby's	Active extraction	Results, lot details
Mecum	Active extraction	Results, lot details
Barrett-Jackson	Active extraction	Results, lot details
Bonhams	Active extraction	Results, lot details
Gooding & Company	Active extraction	Results, lot details
Historics UK	Active extraction	Results, lot details
GAA Classics	Active extraction	Results, lot details
PCarMarket	Active extraction	Listings, results
Collecting Cars	Active extraction	Listings, results
VictoryLap	Active extraction	Listings, results
BH Auction	Active extraction	Results
Hagerty Marketplace	Active extraction	Listings, valuations
eBay Motors	Active extraction	Listings, sold items

Classifieds & Marketplaces (8+ platforms)

Platform	Coverage	Data Depth
Facebook Marketplace	5 dedicated functions	Listings, seller profiles, pricing
Craigslist	3 dedicated functions	Listings, regional coverage
KSL Classifieds	Daily scraping	Listings, pricing
ClassicCars.com	Active extraction	Listings, dealer inventory
Blocket (Sweden)	Active extraction	International listings
Leboncoin (France)	Active extraction	International listings
TheSamba	Active extraction	VW/Porsche specialist
Barn Finds	Active extraction	Discovery/project vehicles

Reference & Enrichment

Source	Data Type
NHTSA VIN Decoder	Factory specifications, recalls
Hagerty Valuation	Editorial price guides
Wayback Machine	Historical listings recovery
Forum build threads	Owner modification history
Service manuals	Technical specifications
Parts catalogs	Component identification

Community & Social

Platform	Data Type
Instagram	Dealer/builder profiles
Twitter/X	Market sentiment
Telegram	Work intake, technician coordination
Forums (Rennlist, etc.)	Build threads, owner history

7.2 Depth of Data Per Vehicle

For a well-covered vehicle, Nuke can provide:

Data Category	Fields	Source
Identity	VIN, year, make, model, trim, body style	VIN decode + observations
Specifications	Engine, transmission, drivetrain, color, interior	VIN decode + listing data
Images	Exterior, interior, engine, undercarriage, detail	Listings (avg 37 images/vehicle)
Pricing History	Every asking price, bid, hammer price, time-series	Multi-platform observations
Market Position	Current value, confidence interval, trend, comps	Valuation engine
Auction Intelligence	Bid curves, comment sentiment, reserve status, watchers	Real-time tracking
Listing History	Every listing across all platforms, days on market	Cross-platform dedup
Condition Signals	Description analysis, photo assessment, owner claims	AI extraction
Provenance	Source URL, extraction timestamp, confidence score	Observation model
Rarity		

Data Category	Fields	Source
	Production numbers, survival estimates, collector demand	Production data + analytics

Average images per vehicle: 37. Compare this to any competitor API that might return 3-5 stock photos.

8. Platform Metrics — Current State

8.1 Database Scale

Metric	Value	Notes
Total Vehicles	**812,913**	Growing ~22K/day
Total Images	**30,373,552**	~37 per vehicle average
Total Observations	**617,385**	New observation system
Legacy Auction Comments	**11,095,537**	BaT historical data
Computed Valuations	**507,071**	62.4% coverage
BaT Identity Seeds	**497,845**	Unique user profiles
Organizations	**2,553**	Dealers, shops, builders
Active Sources	**517**	Registered extraction sources
BaT Listings Indexed	**132,630**	With full comment data
AI Analyses Complete	**127,166**	Comment + description analysis

8.2 Pipeline Throughput

Metric	Value
Vehicles created (last 24h)	**22,775**
Import queue — pending	30,118
Import queue — processing	92
Import queue — complete	163,836
Import queue — failed	1,117 (0.57% failure rate)
GitHub Actions workflows	32 automated pipelines
Edge functions (active)	312
Edge functions (archived)	264

8.3 Data Quality

Metric	Value
Valuation coverage	62.4% of all vehicles
Content deduplication	SHA-256 hash on every observation
Vehicle deduplication	VIN + year/make/model matching
Org deduplication	Name/website/phone normalization + Jaccard similarity
Field-level sourcing	Every field edit tracked with source + confidence
Confidence scoring	4-tier: verified (>=0.95), high (>=0.85), medium (>=0.40), low (<0.40)

8.4 Growth Trajectory

At current ingestion rates: - **22,775 vehicles/day** = ~683K vehicles/month - Targeting 1M vehicles within weeks - At steady state, ~8.2M vehicles/year gross ingest (before dedup) - Observation volume growing as new sources are onboarded

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9. The Observation Model

9.1 Architecture

The observation model is the technical foundation of Nuke's data advantage. It consists of four tables:

observation_sources

The registry of all data sources. Each source has: - `slug` — unique identifier (e.g., `bring_a_trailer`, `facebook_marketplace`) - `source_category` — auction, forum, social_media, marketplace, registry, shop, owner, documentation - `base_trust_score` — baseline confidence for this source (e.g., BaT = 0.75) - `supported_observations` — array of observation kinds this source can produce

vehicle_observations

The core event store. Each observation is: - `vehicle_id` — resolved vehicle reference - `source_slug` — which source produced this observation - `kind` — what type: comment, bid, listing, mileage_reading, sale_result, condition_report - `structured_data` — JSONB payload with type-specific fields - `content_text` — raw text content - `content_hash` — SHA-256 for deduplication - `confidence_score` — computed trust score - `confidence_factors` — breakdown of how confidence was calculated - `observed_at` — when the observation was made (not when it was ingested) - `source_url` — link back to original source

observation_extractors

Configuration for how to extract observations from each source. Defines extraction strategies, field mappings, and processing rules.

observation_discoveries

AI-derived insights from observations. When the system analyzes a set of observations, discoveries are generated — trend identification, anomaly detection, condition assessment.

9.2 Confidence Calculation

Every observation's confidence score is computed as:

```
confidence = base_trust_score(source)
           + 0.10 if vehicle_match_confidence >= 0.95
           + 0.05 if source_url is present
           + 0.05 if content_text > 100 characters
           [capped at 1.0]
```

Vehicle match confidence varies by resolution method: - VIN match: 0.99 - URL match (via external_listings): 0.95 - Fuzzy year/make match: 0.60

Confidence levels: - **Verified** (>= 0.95): VIN-decoded or multi-source corroborated - **High** (>= 0.85): Strong single-source with URL attribution - **Medium** (>= 0.40): Reasonable but uncorroborated - **Low** (< 0.40): Uncertain source or weak match

9.3 Why This Matters

Traditional data providers merge conflicting data and present a single "answer." This destroys information. When KBB says a car is worth $28,500, you can't tell whether that's based on 50 comparable transactions or 2. You can't tell if the transactions are from last week or last quarter. You can't tell if they included the exact trim and options package.

With Nuke's observation model, the API consumer gets: 1. **Every observation** — not a summary, the raw signals 2. **Source attribution** — where each signal came from 3. **Confidence scoring** — how much to trust each signal 4. **Temporal ordering** — the full history, not a point-in-time snapshot

This lets sophisticated consumers (lenders, insurers, funds) build their own models on top of Nuke's observations. They don't need our valuation engine — though we offer one — they need our observations as training data, ground truth, and real-time signal.

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10. Confidence & Provenance — The Real Product

10.1 Field-Level Source Tracking

Nuke tracks provenance at the individual field level via the `vehicle_field_sources` table:

Column	Purpose
`vehicle_id`	Which vehicle
`field_name`	Which field (year, make, model, sale_price, etc.)
`field_value`	The value
`source_type`	How it was obtained: user_input, auction_extraction, ai_extraction, vin_decode
`source_url`	Original source URL
`confidence_score`	0-100 confidence in this value
`user_id`	Who provided it (if manual)
`verified_by`	Who verified it
`verified_at`	When it was verified

Every field edit creates an entry in this table and a corresponding `timeline_event`, maintaining a complete audit trail.

This is what "drills down to the source" means in practice. When a Nuke API response says `year: 1970, year_confidence: 9, year_source: "vin_decode"`, the consumer knows exactly where that data came from and how much to trust it.

10.2 The Provenance Advantage

No other automotive data API exposes this level of transparency. The implications:

For lenders: “We valued this vehicle at $45,000 based on 23 comparable transactions from 3 platforms over the last 90 days, with a confidence score of 8.2/10.” This is auditable. This satisfies examiners.

For insurers: “The replacement value is $52,000-$68,000 based on recent auction results for this make/model/year in Condition 2-3, sourced from BaT and RM Sotheby’s.” This is defensible.

For dealers: “This vehicle is priced 12% below market based on 7 comparable sales. The last similar vehicle on BaT sold for $38,500 with 34 bids.” This is actionable.

For AI/ML teams: “Here are 500,000 labeled vehicle sale observations with source, timestamp, confidence, and 37 images per vehicle.” This is training data.

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11. The Valuation Engine

11.1 Architecture

Nuke's valuation engine (`compute-vehicle-valuation`) is a multi-signal weighted model. Unlike editorial valuations (Hagerty, KBB), it is computed from observed transaction data.

Signal Weights by Price Tier

The engine adapts its methodology based on the vehicle's price tier, recognizing that different factors matter at different price points:

Signal	Budget (<$15K)	Mainstream (<$50K)	Enthusiast (<$150K)	Collector (<$500K)	Trophy ($500K+)
Comparables	0.45	0.40	0.30	0.25	0.20
Condition	0.18	0.15	0.12	0.10	0.08
Rarity	0.02	0.05	0.10	0.15	0.18
Sentiment	0.05	0.08	0.12	0.13	0.12
Bid Curve	0.10	0.10	0.08	0.05	0.04
Market Trend	0.10	0.10	0.08	0.07	0.06
Survival Rate	0.02	0.04	0.08	0.10	0.12
Originality	0.08	0.08	0.12	0.15	0.20

Key insight: For budget/mainstream vehicles, comparables dominate (they're common, data is abundant). For collector/trophy vehicles, rarity, originality, and survival rate matter more — and Nuke accounts for this automatically.

Confidence Intervals

Price Tier	Confidence Interval
Budget	±8%
Mainstream	±12%
Enthusiast	±18%
Collector	±25%
Trophy	±35%

Wider intervals at higher price points reflect the reality that trophy vehicles are unique — each one is a market of one.

11.2 Deal Score

Every listed vehicle gets a deal score indicating whether it's priced above or below market:

```
deal_score = ((estimated_value - asking_price) / estimated_value) ×
freshness_decay
```

Freshness decay ensures that old listings are penalized: - <24 hours: 100% weight - 1-7 days: 90% - 7-14 days: 70% - 14-30 days: 50% - >30 days: 30%

Labels range from `plus_3` (exceptional deal) through `fair` to `minus_3` (significantly overpriced).

11.3 Heat Score

A 0-100 composite measuring current market interest:

Factor	Points
Active auction	+30
Recent sale (<7 days)	+25
Strong deal score	+15
High bid velocity	+10
Community buzz (comments)	+10
Ultra-rare vehicle	+10
New listing (<48h)	+5

Labels: cold / warm / hot / fire / volcanic

11.4 Record Proximity Effect

When a vehicle's estimated value approaches the record price for its segment, the engine automatically widens the confidence interval. This prevents false precision at the edge of known data and signals to consumers that they're in uncharted territory.

---

12. Source Coverage

12.1 Platform Integration Matrix

Platform	Listings	Sold Prices	Images	Comments	Bids	Sentiment	Seller Data
Bring a Trailer	Yes	Yes	Yes	Yes (11M+)	Yes	Yes	Yes
Cars & Bids	Yes	Yes	Yes	Yes	Yes	Yes	Partial
RM Sotheby's	Yes	Yes	Yes	No	No	No	No
Mecum	Yes	Yes	Yes	No	No	No	No
Barrett-Jackson	Yes	Yes	Yes	No	No	No	No
Bonhams	Yes	Yes	Yes	No	No	No	No
Gooding	Yes	Yes	Yes	No	No	No	No
eBay Motors	Yes	Yes	Yes	No	No	No	Yes
Facebook Marketplace	Yes	No	Yes	No	No	No	Yes
Craigslist	Yes	No	Yes	No	No	No	Partial
Hagerty	Yes	Partial	Yes	No	No	No	No
Collecting Cars	Yes	Yes	Yes	No	No	No	No
PCarMarket	Yes	Yes	Yes	No	No	No	No

12.2 Coverage by Vehicle Segment

Segment	Estimated Market Size	Nuke Coverage	Primary Sources
Classic American (pre-1975)	~2M vehicles	Strong	BaT, Mecum, Barrett-Jackson, Craigslist
European Sports/GT	~500K vehicles	Strong	BaT, RM Sotheby's, Bonhams, Collecting Cars
Japanese Classics	~300K vehicles	Moderate	BaT, C&B, Facebook, forums
Trucks & SUVs	~5M vehicles	Growing	BaT, Craigslist, Facebook, KSL
Modern Performance	~2M vehicles	Moderate	C&B, eBay, Facebook
Exotics/ Supercars	~200K vehicles	Moderate	RM Sotheby's, Bonhams, Gooding
Barn Finds/ Projects	~1M vehicles	Growing	Barn Finds, Craigslist, Facebook

12.3 International Coverage

Region	Sources	Status
North America	20+ platforms	Primary focus
United Kingdom	Historics UK, Bonhams	Active
France	Leboncoin	Active
Sweden	Blocket	Active
Rest of Europe	Collecting Cars, RM Sotheby's EU	Partial
Asia/Pacific	Limited	Future

13. Pipeline & Ingestion Architecture

13.1 Data Flow

```
[External Sources]
    ↓
[Source-Specific Extractors] (59 dedicated functions)
    ↓
[Observation Intake] (ingest-observation)
    ↓ SHA-256 dedup check
    ↓ Vehicle resolution (VIN → URL → fuzzy)
    ↓ Confidence scoring
    ↓
[vehicle_observations] ← immutable event store
    ↓
[Vehicle Enrichment]
    ├── VIN decode → specs population
    ├── Image analysis → categorization, quality scoring
    ├── Comment analysis → sentiment, themes
    ├── Valuation compute → estimates, deal scores, heat
    └── Discovery engine → AI insights
    ↓
[vehicles] ← current-state materialized view
    ↓
[API Layer] → consumers
```

13.2 Extraction Methods

Method	Technology	Use Case	Functions
HTTP Scraping	Deno fetch + cheerio	Simple listing pages	~30 extractors
Browser Automation	Playwright	JavaScript-heavy sites, anti-bot	~10 extractors
AI Extraction	Claude/GPT + raw HTML	Unstructured/novel layouts	3 generic extractors
Local LLM	Ollama	Cost-sensitive extraction	1 extractor
Firecrawl	API-based scraping service	Rate-limited or complex sites	~5 extractors
Feed Polling	RSS/API monitoring	Continuous monitoring	1 orchestrator

13.3 Orchestration

System	Purpose
Ralph Wiggum Coordinator	System health monitoring, queue management, failure routing
Agent Orchestrator	Multi-agent task coordination
Pipeline Orchestrator (GHA)	Scheduled extraction across all sources
Crawler Scheduler	Adaptive crawling based on source update frequency
Continuous Queue Processor	Background processing of import queue

13.4 Resilience

- **Retry logic**: Exponential backoff on extraction failures

- **Circuit breaker**: Per-domain failure tracking (e.g., Barrett-Jackson: 250 failures → paused)
- **Dedup protection**: Content hashing prevents duplicate ingestion even on re-crawl
- **Queue persistence**: All jobs tracked in `import_queue` with status lifecycle
- **Failure rate**: 0.57% (1,117 failed / 195,071 total)

---

14. Current API Surface

14.1 Production Endpoints

Core Vehicle API

Endpoint	Method	Auth	Purpose
`/api-v1-vehicles`	GET	JWT/API Key	List vehicles with pagination
`/api-v1-vehicles/{id}`	GET	JWT/API Key	Get single vehicle
`/api-v1-vehicles`	POST	JWT/API Key	Create vehicle
`/api-v1-vehicles/{id}`	PATCH	JWT/API Key	Update vehicle
`/api-v1-vehicles/{id}`	DELETE	JWT/API Key	Archive vehicle

Observations API

Endpoint	Method	Auth	Purpose
`/api-v1-observations`	GET	JWT/API Key	List observations (filter by vehicle_id, vin, kind)
`/api-v1-observations`	POST	JWT/API Key	Create observation (auto-resolves VIN)

Batch API

Endpoint	Method	Auth	Purpose
`/api-v1-batch`	POST	JWT/API Key	Bulk import (1000 vehicles, 100 obs/vehicle)

Search

Endpoint	Method	Auth	Purpose
`/universal-search`	POST	Optional	Smart search (VIN, URL, text detection)
`/search`	POST	Optional	Advanced FTS + fuzzy search

API Management

Endpoint	Method	Auth	Purpose
`/api-keys-manage`	GET/POST/DELETE	JWT only	API key lifecycle
`/webhooks-manage`	GET/POST/PATCH/DELETE	JWT/API Key	Webhook CRUD + secret rotation

Platform

Endpoint	Method	Auth	Purpose
`/db-stats`	GET	Service role	Database statistics
`/platform-status`	GET	None	Platform configuration

14.2 Authentication

Two methods, both supported on all core endpoints:

JWT Bearer Token: Standard Supabase JWT (HS256, 1hr expiry)

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

API Key: Prefixed key, SHA-256 hashed for storage

```
X-API-Key: nk_live_a1b2c3d4e5f6g7h8...
```

14.3 OpenAPI Specification

A complete OpenAPI 3.0.3 specification exists at `/docs/api/openapi.yaml` (981 lines), covering vehicles, observations, batch, and webhooks endpoints. This is ready for documentation generation.

14.4 What's Not Yet Exposed

The following data exists in the database but has no public API endpoint:

Data	Table	Current Access
Auction results & bid history	external_listings	Internal only
Market valuations	nuke_estimates	Internal only
Market comparables	Computed on demand	Internal only
Market trends & indexes	Computed on demand	Internal only
Production/ rarity data	vehicle_production_data	Internal only
Dealer/org profiles	businesses	Internal only
Dealer inventory	business_vehicle_fleet	Internal only
Auction comments & sentiment	auction_comments + analysis	Internal only
Image analysis results	vehicle_images metadata	Internal only

This is the single biggest gap. Nuke's strongest data is locked behind internal functions.

15. The SDK

15.1 Current State

Package: `@nuke/sdk` v1.0.0 Runtime: Node.js 18+ Build: tsup (CJS + ESM + DTS)

```
import Nuke from '@nuke/sdk';

const nuke = new Nuke('nk_live_...');

// Vehicles
const vehicle = await nuke.vehicles.create({ year: 1970, make:
        'Porsche', model: '911S' });
const vehicles = await nuke.vehicles.list({ page: 1, limit: 25 });
const all = nuke.vehicles.listAll(); // async generator, auto-
        pagination

// Observations
const obs = await nuke.observations.create({
  vehicle_id: '...',
  source_type: 'manual',
  data_type: 'price',
  value: '45000',
});

// Batch
const result = await nuke.batch.ingest({
  vehicles: [...],
  options: { match_by: 'vin', skip_duplicates: true }
});

// Webhooks
const hook = await nuke.webhooks.create({
  url: 'https://...',
  events: ['vehicle.created', 'observation.created']
});

// Webhook signature verification (Stripe-compatible)
const event = nuke.webhooks.constructEvent(payload, signature,
        secret);
```

15.2 Error Hierarchy

```
NukeError
├── NukeAPIError (base for all API errors)
│   ├── NukeAuthenticationError (401)
│   ├── NukeRateLimitError (429)
│   ├── NukeValidationError (400)
│   └── NukeNotFoundError (404)
```

15.3 Design Patterns

The SDK follows Stripe/Plaid conventions: - Resource namespaces (`nuke.vehicles`, `nuke.observations`) - `listAll()` async generators for auto-pagination - Webhook signature verification with timing-safe comparison - Batch ingestion with chunking and progress callbacks - Typed responses with full TypeScript definitions

15.4 What's Missing

- **Not published to npm** — exists in `tools/nuke-sdk/` but not released
- **No valuation endpoints** — because the API doesn't expose them yet
- **No market data endpoints** — same reason
- **No auction data endpoints** — same
- **No documentation site** — no hosted docs

---

16. Webhooks & Event System

16.1 Current Implementation

Nuke has a production-ready webhook system modeled on Stripe's:

Supported Events: - `vehicle.created` - `vehicle.updated` - `vehicle.deleted` - `observation.created` - `document.uploaded` - `import.completed` - `*` (wildcard)

Security: - Secrets: `whsec_<64-hex-chars>` (shown once on creation) - Signatures: HMAC-SHA256 in format `t={timestamp},v1={signature}` - Dual headers: `Nuke-Signature` and `Stripe-Signature` for interop - Secret rotation support

Reliability: - Exponential backoff: 1min, 5min, 30min, 2hr, 8hr (max 5 retries) - Delivery tracking in `webhook_deliveries` table - Per-endpoint delivery stats (total/failed) - Scheduler-callable retry for failed deliveries

16.2 What Should Be Added

For an API-first business, the webhook system needs more events:

Event Category	Events Needed
Valuation	`valuation.computed`, `valuation.changed`, `deal_score.changed`
Auction	`auction.started`, `auction.bid_placed`, `auction.ending_soon`, `auction.completed`
Market	`market.trend_alert`, `market.record_price`
Listing	`listing.appeared`, `listing.price_changed`, `listing.sold`, `listing.expired`

These events represent the real-time data streams that premium API consumers will pay the most for.

---

17. The Target API Surface

17.1 Proposed Endpoint Structure

```
# Vehicle Data
GET  /v1/vehicles/{vin}                    → Identity, specs,
images, current status
GET  /v1/vehicles/{vin}/history            → Full observation
timeline
GET  /v1/vehicles/{vin}/images             → All images with
categories and metadata
GET  /v1/vehicles/{vin}/valuations         → Current value,
confidence, range, comps
GET  /v1/vehicles/{vin}/listings           → All listings (active +
historical, cross-platform)
GET  /v1/vehicles/{vin}/auction            → Auction results, bids,
comments, sentiment
GET  /v1/vehicles/{vin}/similar            → Similar vehicles
currently available

# Market Data
GET  /v1/market/comps                      → Comparable sales
(filter by make/model/year/condition)
GET  /v1/market/trends                     → Price trends, volume,
days-on-market
GET  /v1/market/indexes                    → Segment indexes
(muscle, JDM, euro, exotic, truck)
GET  /v1/market/records                    → Record sales by segment
GET  /v1/market/alerts                     → Subscribe to market
conditions

# Search & Discovery
GET  /v1/search                            → Unified search (VIN,
text, filters)
GET  /v1/discover                          → Recommendation engine
(similar to, trending, undervalued)
GET  /v1/feed                              → Chronological feed of
new observations

# Dealer / Organization Data
GET  /v1/dealers/{id}                      → Dealer profile,
reputation, capabilities
GET  /v1/dealers/{id}/inventory            → Current stock
GET  /v1/dealers/search                    → Find dealers by
location, specialty, rating

# Data Ingestion
POST /v1/vehicles                          → Create vehicle
POST /v1/observations                      → Submit observation
POST /v1/batch                             → Bulk import
```

```
# Management
POST /v1/auth/keys                          → Create API key
GET  /v1/auth/keys                          → List API keys
POST /v1/webhooks                           → Create webhook
GET  /v1/usage                              → Current usage and quota
```

17.2 Design Principles

1. **VIN-first routing.** The VIN is the natural primary key for the automotive industry. Every vehicle endpoint should accept a VIN as the identifier.

2. **Every field carries provenance.** No response should contain a value without a corresponding source and confidence indicator.

3. **Observations are first-class.** The `/history` endpoint returns raw observations, not a summary. Consumers can filter by source, kind, date range, and confidence level.

4. **Market data is derived, not editorial.** Every market endpoint should link back to the observations that inform it.

5. **Real-time over periodic.** Auction data should be live. Valuations should update when new observations arrive, not on a schedule.

---

18. API Design Philosophy

18.1 Learning from the Best

The APIs that developers love — Stripe, Twilio, Plaid — share common traits:

Stripe: - Consistent resource patterns (CRUD on everything) - Exhaustive documentation with examples in every language - Webhook-first architecture for async events - Idempotency keys for safe retries - Rich error objects with actionable messages

Twilio: - Every resource has a unique SID - Pagination via next_page_uri (cursor-based) - Versioned URLs (2010-04-01 is still supported) - Subresources (Calls → Recordings → Transcriptions)

Plaid: - Clean mental model (Link → Access Token → Data) - Sandbox environment with predictable test data - Error codes that tell you exactly what went wrong - Webhooks for async data availability

18.2 Nuke API Principles

1. Source-attributed responses. Every value in every response includes provenance metadata. This is not optional or an add-on — it's the core product.

2. Confidence as a first-class concept. Confidence scores appear at the field level, the observation level, and the valuation level. API consumers can filter by minimum confidence.

3. Temporal queries. Every endpoint supports `as_of` parameters for point-in-time queries. "What did we know about this vehicle on January 15?" is a valid query.

4. Cursor-based pagination. No offset/limit — use cursors for stable, performant pagination across large datasets.

5. Consistent error taxonomy.

```
{
  "error": {
    "type": "invalid_request",
    "code": "vehicle_not_found",
    "message": "No vehicle found with VIN 1GCEK19K5RE123456",
    "param": "vin",
    "doc_url": "https://docs.nuke.dev/errors/vehicle_not_found"
  }
}
```

6. Idempotency. All POST endpoints accept an `Idempotency-Key` header for safe retries.

7. Versioning. URL-based versioning (`/v1/`, `/v2/`) with minimum 24-month deprecation windows.

---

19. Developer Experience Roadmap

19.1 Documentation

Component	Status	Priority
OpenAPI 3.0 spec	Exists (981 lines)	Ship as-is, expand
Interactive API reference	Not built	P0 — needed for launch
Getting Started guide	Not written	P0
Authentication guide	Not written	P0
Webhook integration guide	Not written	P1
SDK documentation	Not written	P1
Example applications	Not built	P2
Changelog	Not started	P1
Status page	Not built	P1

19.2 Developer Portal

Components needed: - **Dashboard**: API key management, usage metrics, billing - **API Reference**: Auto-generated from OpenAPI spec (Stoplight, Redocly, or custom) - **Sandbox**: Test environment with seeded data - **Playground**: Interactive API explorer (try before you sign up) - **Webhook Testing**: Webhook delivery logs and replay

19.3 SDK Expansion

SDK	Status	Priority
TypeScript/Node.js	Built, unpublished	P0 — publish to npm
Python	Not built	P1 — ML/data science audience
Go	Not built	P2 — infrastructure/backend audience
REST (curl examples)	In OpenAPI spec	P0 — include in docs

19.4 MCP Server

Nuke already has an MCP (Model Context Protocol) server, positioning it as an AI-native data source. This is forward-looking — as AI agents become primary API consumers, having an MCP integration means Nuke data is available in Claude, Cursor, and other LLM-powered tools without custom integration.

---

20. Pricing Strategy

20.1 Current Pricing

Product	Price	Unit
API Access (monthly)	$29.99/mo	1,000 image analyses
Prepaid 100	$4.99	100 analyses
Prepaid 500	$19.99	500 analyses
Prepaid 1,000	$34.99	1,000 analyses
DealerScan 100	$20.00	100 document extractions
DealerScan 500	$90.00	500 extractions
DealerScan 1,000	$160.00	1,000 extractions

Problem: This pricing is product-feature-based (image analysis, document extraction), not data-access-based. For an API-first positioning, the pricing model needs to reflect data consumption.

20.2 Proposed Pricing Model

Tier Structure

Tier	Monthly Price	Included	Overage	Target Customer
Build	Free	500 VIN lookups, 100 observations	—	Hobbyists, evaluation
Startup	$149/mo	5,000 VIN lookups, 1,000 observations, basic market data	$0.05/ lookup	Early-stage apps
Scale	$499/mo	25,000 VIN lookups, 10,000 observations, full market data, webhooks	$0.03/ lookup	Growing businesses
Pro	$1,499/mo	100,000 VIN lookups, unlimited observations, real-time auction feed, priority support	$0.02/ lookup	Dealers, lenders
Enterprise	Custom	Unlimited, firehose access, SLA, dedicated support, custom endpoints	Negotiated	Insurers, funds, OEMs

Premium Add-Ons

Add-On	Price	Description
Real-time auction feed	$299/mo	Live bid updates, soft-close alerts
Sentiment analysis	$199/mo	Comment-derived sentiment on auctions
Bulk historical export	$0.01/record	One-time data pulls for model training
Dedicated IP / VPN peering	$500/mo	Enterprise network integration

Rationale

1. **VIN lookup as the unit of value.** A VIN lookup is the atomic action — it's how customers think about their usage. Everything else (observations, valuations, history) is delivered as part of the VIN response.

2. **Free tier for adoption.** 500 lookups/month is enough to build a prototype, not enough to run a business. This drives self-serve signups.

3. **Market data as a tier gate.** Basic VIN data is available at all tiers. Market comps, trends, and indexes unlock at Scale and above. This creates natural upsell pressure.

4. **Real-time as premium.** The auction firehose is the highest-value, highest-cost data stream. It's priced as an add-on to avoid forcing it on customers who don't need it.

20.3 Competitive Pricing Context

Competitor	Pricing	Unit
MarketCheck	$99-$999/mo	5K-100K requests
Black Book	$30K-$200K/yr	Enterprise contract
KBB API	$50K-$500K/yr	Enterprise contract
J.D. Power	Custom enterprise	Bundled
Drivly	~$0.05-$0.15/call	Per-call
CarFax (history)	~$40/report	Per-report
Edmunds	Free-custom	Per-call

Nuke's proposed pricing is competitive with MarketCheck at the low end and dramatically cheaper than KBB/Black Book at the enterprise level — while offering data that neither provides (auction results, provenance, confidence scoring).

---

21. Customer Segments

21.1 Primary Segments

Segment 1: Dealer Tools & DMS Platforms

Who: Companies building dealer management systems, pricing tools, inventory management, and CRM platforms. **What they need:** Real-time market pricing, comparable sales, competitive inventory analysis, automated appraisal tools. **Why Nuke:** Transaction-based pricing (not estimates), cross-platform visibility (see what competitors are listing), real-time data. **Example use:** Auto-populate trade-in values based on actual comparable sales, alert when a vehicle is priced below market. **Revenue potential:** $1,000-$10,000/mo per customer. 500+ potential customers in US alone.

Segment 2: Automotive Lenders & Fintech

Who: Banks, credit unions, fintech companies offering auto loans, refinancing, or vehicle-backed lending. **What they need:** Defensible valuations for loan-to-value calculations, audit trails for regulators, real-time value updates for portfolio monitoring. **Why Nuke:** Source-attributed valuations satisfy examiner requirements. Confidence intervals enable risk calibration. Transaction data (not guidebook estimates) reduces model risk. **Example use:** Real-time LTV monitoring on a portfolio of 50,000 auto loans, with alerts when collateral value drops below threshold. **Revenue potential:** $5,000-$50,000/mo per customer. 200+ potential customers.

Segment 3: Insurance & Insurtech

Who: Specialty vehicle insurers (Hagerty competitors), agreed-value policy providers, claims adjusters. **What they need:** Accurate replacement values for specialty/classic vehicles, market trend data for portfolio risk, claims validation. **Why Nuke:** Covers classic/enthusiast vehicles that KBB/Black Book don't. Transaction-based replacement values. Confidence intervals for underwriting

models. **Example use:** Policy issuance with automated agreed-value recommendation based on recent comparable sales. **Revenue potential:** $5,000-$100,000/mo per customer. 100+ potential customers.

Segment 4: Marketplace & Auction Platforms

Who: Online vehicle marketplaces, auction platforms, classified sites. **What they need:** Instant pricing guidance for sellers, market comparison for buyers, listing enrichment. **Why Nuke:** Cross-platform pricing data. Observation history adds value to listings. Market position indicators (deal score) drive buyer engagement. **Example use:** "This vehicle is priced 8% below similar vehicles on other platforms" displayed on a listing page. **Revenue potential:** $2,000-$20,000/mo per customer. 50+ potential customers.

Segment 5: Data Science & AI/ML Teams

Who: Teams building valuation models, price prediction, market analysis, or training automotive AI. **What they need:** Labeled datasets with images, prices, specs, and transaction outcomes. Clean, structured, large-scale data. **Why Nuke:** 812K vehicles, 30M images, 500K+ valuations with confidence labels, source attribution. This is training data, not an API to call in production — it's the foundation of their models. **Example use:** Train a vehicle condition assessment model on 30M labeled images with known sale prices. **Revenue potential:** $10,000-$100,000 per dataset export. 100+ potential customers.

Segment 6: Investment & Collector Funds

Who: Automotive investment funds, fractional ownership platforms, collector vehicle portfolio managers. **What they need:** Market indexes, trend analysis, comparable transaction data, portfolio valuation. **Why Nuke:** Segment-level indexes (muscle, JDM, euro, exotic). Record price tracking. Real-time auction monitoring. Historical trend data. **Example use:** Monthly mark-to-market on a $50M collector vehicle portfolio using Nuke's comparable transaction data. **Revenue potential:** $10,000-$50,000/mo per customer. 50+ potential customers.

21.2 Market Sizing

Segment	Customers	Avg Revenue	Annual Revenue Potential
Dealer Tools	500	$5K/mo	$30M
Lenders/Fintech	200	$15K/mo	$36M
Insurance	100	$20K/mo	$24M
Marketplaces	50	$10K/mo	$6M
Data Science	100	$50K/yr	$5M
Investment Funds	50	$25K/mo	$15M
Total	**1,000**		**$116M ARR potential**

This is the addressable market at full product maturity. Near-term (12-month) target: 20-50 paying customers, $500K-$2M ARR.

---

22. Go-to-Market

22.1 Phase 1: Developer Adoption (Months 1-3)

Goal: 500 free-tier signups, 20 paying customers.

Actions: 1. Ship the public API with the target endpoint surface 2. Publish the TypeScript SDK to npm 3. Launch docs site (docs.nuke.dev) 4. Launch on Hacker News ("Show HN: The automotive data API with source provenance on every field") 5. Post on Product Hunt 6. Submit to API directories (RapidAPI, API Layer, ProgrammableWeb) 7. Write technical blog posts: - "Why we built an observation model for automotive data" - "How we track confidence scores across 25 auction platforms" - "Building a vehicle valuation engine from transaction data"

Metrics: Free tier signups, API calls/day, SDK downloads, documentation page views.

22.2 Phase 2: Design Partners (Months 3-6)

Goal: 5 enterprise design partners, product-market fit validation.

Actions: 1. Identify 20 targets across dealer tools, lending, and insurance segments 2. Offer free Pro-tier access for 90 days in exchange for feedback 3. Build custom integrations for 3-5 design partners 4. Iterate on API surface based on partner feedback 5. Develop case studies from partner implementations

Metrics: Design partner NPS, API consumption growth, feature requests logged, conversion to paid.

22.3 Phase 3: Scale (Months 6-12)

Goal: 50+ paying customers, $500K+ ARR.

Actions: 1. Launch self-serve billing (Stripe integration already exists) 2. Hire developer relations / developer advocate 3. Conference presence (NADA, DealerSocket, auto fintech events) 4. Partner with DMS platforms for embedded integrations 5. Launch Python SDK for data science audience 6. Introduce enterprise SLAs and dedicated support

Metrics: ARR, customer count by tier, churn rate, API uptime, support ticket volume.

22.4 Positioning Statement

For developers and data teams building automotive applications **Who** need reliable, traceable vehicle data across the full market **Nuke** is the automotive data API **That** provides source-attributed, confidence-scored vehicle data from 25+ platforms in a single, developer-friendly interface **Unlike** KBB, Black Book, or Hagerty, which deliver opaque editorial estimates **Nuke** shows you exactly where every data point came from and how much to trust it.

---

23. Competitive Positioning Matrix

23.1 Feature Comparison

Capability	Nuke	KBB	Black Book	Hagerty	MarketCheck	Classic.com	Drivly
VIN Decode	Yes	Yes	Yes	No	Yes	No	Yes
New Car Pricing	No	Yes	Yes	No	Partial	No	Yes
Used Car Valuation	Yes	Yes	Yes	Partial	No	Partial	Partial
Classic/ Enthusiast	**Yes**	No	No	**Yes**	No	**Yes**	No
Auction Results	**Yes**	No	No	Partial	No	**Yes**	No
Real-time Auctions	**Yes**	No	No	No	No	No	No
Listing Aggregation	**Yes**	No	No	No	**Yes**	No	No
Transaction History	**Yes**	No	Partial	No	No	Partial	No
Source Provenance	**Yes**	No	No	No	No	No	No
Confidence Scoring	**Yes**	No	No	No	No	No	No
Field-level Sources	**Yes**	No	No	No	No	No	No
Observation Model	**Yes**	No	No	No	No	No	No
Comment Sentiment	**Yes**	No	No	No	No	No	No
Images (avg/ vehicle)	37	3-5	0	2-3	5-10	3-5	0
	Yes	No	No	No	**Yes**	Partial	**Yes**

Capability	Nuke	KBB	Black Book	Hagerty	MarketCheck	Classic.com	Drivly
Self-serve API							
Webhooks	**Yes**	No	No	No	No	No	No
TypeScript SDK	**Yes**	No	No	No	No	No	No

23.2 Positioning Map

```
                    Editorial / Opaque
                          ↑
                          |
            KBB ●         |       ● Black Book
                          |
            Hagerty ●     |       ● J.D. Power
                          |
      Classic ←-----------+-----------→ Modern
                          |
        Classic.com ●     |       ● MarketCheck
                          |
            Nuke ●        |       ● Drivly
                          |
                          ↓
                    Transaction / Transparent
```

Nuke occupies a unique position: **broad vehicle coverage** (classic through modern) with **transaction-level transparency**. No competitor occupies this quadrant.

24. Technical Moat Analysis

24.1 What's Hard to Replicate

1. The extraction pipeline. 312 active functions across 25+ platforms, each with platform-specific logic for handling authentication, pagination, anti-bot measures, and data normalization. This represents 18+ months of iteration. A competitor starting from scratch would need 12-18 months to reach equivalent coverage.

2. The historical dataset. 812K vehicles, 30M images, 11M auction comments, 617K observations. This is accumulated over time — it cannot be created on demand. Historical auction results (especially pre-2024) are increasingly difficult to scrape as platforms archive older content.

3. The observation model. The architectural decision to use an append-only observation model rather than a mutable vehicle record is fundamental. Retrofitting this onto a traditional database design is a full rewrite. Nuke was built observation-first.

4. The confidence framework. Source trust scores, field-level provenance, and tiered confidence are woven into every layer of the system — extraction, storage, API, and SDK. This is not a feature that can be bolted on.

5. The valuation engine. An 8-signal weighted model with tier-dependent weights, deal scoring with freshness decay, heat scoring, and record proximity effects. This was calibrated on 500K+ valuations and is continuously improving.

24.2 What Could Be Replicated

1. VIN decoding. NHTSA data is free. Any competitor can build a VIN decoder.

2. Basic listing scraping. Individual platform scrapers are not defensible. The breadth and reliability of 25+ is.

3. Market comparison. Simple comparable-sales logic is well-understood. The observation-based, confidence-weighted approach is harder.

4. API packaging. REST API design is commodity. The data behind it is not.

24.3 Moat Strength Assessment

Moat Component	Strength	Time to Replicate
Extraction pipeline breadth	Strong	12-18 months
Historical dataset depth	Very Strong	2-3 years (data is time-locked)
Observation architecture	Strong	6-12 months (architecture decision)
Confidence framework	Moderate-Strong	6 months (concept is describable)
Valuation engine	Moderate	3-6 months (methodology is publishable)
Developer experience	Weak (not yet shipped)	2-3 months
Brand / network effects	Weak (early stage)	12-24 months

Conclusion: The strongest moats are the historical dataset (cannot be replicated without time) and the extraction pipeline (breadth + reliability). The weakest areas are developer experience and brand — which is why shipping the public API and docs is the highest priority.

25. Risks & Mitigations

25.1 Technical Risks

Risk	Likelihood	Impact	Mitigation
Platform blocks scraping	High	Medium	Diversified sources (25+), Playwright fallback, proxy rotation. No single source represents >30% of data.
Data quality degradation	Medium	High	Content hashing prevents duplicates. Confidence scoring surfaces low-quality data. Quality monitoring workflows (scrape-quality-monitor).
API uptime	Medium	High	Supabase Edge Functions have built-in redundancy. Need to add dedicated status page and SLA monitoring.
Scale limitations	Low	Medium	Supabase handles well to ~1M rows per table. Beyond that, may need to partition or migrate hot paths.

25.2 Business Risks

Risk	Likelihood	Impact	Mitigation
Incumbents launch similar products	Medium	Medium	KBB/Black Book are structurally unable to offer source transparency (their value is the opaque model). Hagerty could but hasn't. First-mover advantage on provenance.
Legal challenges from platforms	Medium	High	Data is factual (sale prices, listing details) — generally not copyrightable. However, ToS violations could trigger cease-and-desists. Mitigation: API access where available, respectful scraping rates, caching over re-scraping.
Pricing pressure	Low	Medium	Nuke's differentiation is not "cheaper data" but "better data." Provenance and confidence scoring justify premium over commodity alternatives.
Slow enterprise sales cycles	High	Medium	Self-serve tiers generate cash flow while enterprise deals mature. Free tier drives adoption and creates inbound demand.

25.3 Market Risks

Risk	Likelihood	Impact	Mitigation
Market contraction (auto downturn)	Medium	Low-Medium	Data demand is counter-cyclical in some segments (lenders need more data in downturns for risk management).
Platform consolidation	Low	Medium	If BaT, C&B, etc. consolidate, could reduce data diversity. Mitigated by observation model — historical data remains valuable regardless.
AI disruption of valuations	Medium	Low	AI can generate valuations, but it needs training data. Nuke provides the training data. AI is a customer, not a competitor.

26. 90-Day Execution Plan

Month 1: Ship the API

Week	Deliverable
1	Stand up `/v1/vehicles/{vin}` (unified endpoint wrapping existing data)
1	Stand up `/v1/vehicles/{vin}/history` (observations timeline)
2	Stand up `/v1/vehicles/{vin}/valuations` (expose nuke_estimates)
2	Stand up `/v1/vehicles/{vin}/listings` (expose external_listings)
3	Stand up `/v1/market/comps` and `/v1/market/trends`
3	Stand up `/v1/search` (wrap universal-search)
4	Implement usage metering and tier enforcement
4	Publish TypeScript SDK to npm

Month 2: Ship the Developer Experience

Week	Deliverable
5	Launch docs.nuke.dev (auto-generated from OpenAPI + hand-written guides)
5	Build interactive API playground
6	Create sandbox environment with seeded test data
6	Write Getting Started guide, Authentication guide, Webhooks guide
7	Launch self-serve signup and billing (Stripe)
7	Build developer dashboard (API keys, usage, billing)
8	Launch on Hacker News, Product Hunt, API directories

Month 3: First Customers

Week	Deliverable
9	Outreach to 20 target design partners
9	Stand up `/v1/vehicles/{vin}/auction` (bid history, comment sentiment)
10	Build 3 example integrations (dealer pricing tool, lender LTV calculator, market dashboard)
10	Add real-time auction webhooks
11	Iterate based on first customer feedback
11	Stand up `/v1/dealers` endpoints
12	Write first case study
12	Assess: what's working, what needs to change

Key Dependencies

1. **Domain**: Secure `api.nuke.dev` and `docs.nuke.dev`
2. **Hosting**: Decide if API gateway sits in front of Supabase Edge Functions or if Phoenix backend serves the public API
3. **Metering**: Build or buy usage metering (Stripe Billing, Metronome, or custom)
4. **Support**: Set up support channel (Intercom, Discord, or GitHub Discussions)

---

27. Appendix A: Current Database Schema (Key Tables)

vehicles

The central table with 100+ columns:

Column Group	Key Fields
Identity	id (uuid), vin, year, make, model, trim, body_style
Specs	engine, transmission, drivetrain, color, interior_color, mileage
Pricing	sale_price, asking_price, msrp, current_value, winning_bid, high_bid
Confidence	year_confidence, make_confidence, vin_confidence, price_confidence (1-10)
Sources	year_source, make_source, discovery_source, discovery_url, platform_source
Status	is_public, is_for_sale, is_modified, condition_rating
Ownership	user_id, created_at, updated_at

vehicle_observations

Column	Type	Purpose
id	uuid	Primary key
vehicle_id	uuid	FK to vehicles
source_slug	text	FK to observation_sources
kind	text	comment, bid, listing, sale_result, etc.
structured_data	jsonb	Type-specific payload
content_text	text	Raw content
content_hash	text	SHA-256 for dedup
confidence_score	float	0-1 trust score
confidence_factors	jsonb	Score breakdown
observed_at	timestamptz	When the event occurred
source_url	text	Original source

external_listings

Column	Type	Purpose
id	uuid	Primary key
vehicle_id	uuid	FK to vehicles
platform	text	bat, cars_and_bids, ebay_motors, etc.
listing_url	text	Original listing URL
listing_status	text	pending, active, ended, sold, cancelled
current_bid	numeric	Current bid amount
reserve_price	numeric	Reserve (if known)
final_price	numeric	Hammer price
bid_count	integer	Number of bids
view_count	integer	Listing views
start_date	timestamptz	Listing start
end_date	timestamptz	Listing end
sold_at	timestamptz	Sale timestamp

nuke_estimates (valuations)

Column	Type	Purpose
vehicle_id	uuid	FK to vehicles
estimated_value	numeric	Computed market value
confidence_score	float	Valuation confidence
confidence_interval	numeric	± range
comparable_count	integer	Number of comps used
signal_weights	jsonb	Weight breakdown
deal_score	text	plus_3 to minus_3
heat_score	integer	0-100
computed_at	timestamptz	When calculated

vehicle_field_sources

Column	Type	Purpose
vehicle_id	uuid	FK to vehicles
field_name	text	Which field
field_value	text	The value
source_type	text	user_input, auction_extraction, ai_extraction, vin_decode
source_url	text	Original source
confidence_score	integer	0-100
verified_by	uuid	Verifier user

28. Appendix B: Full Endpoint Inventory (Current)

Production Edge Functions by Category

Core API (7 functions)

- api-v1-vehicles
- api-v1-observations
- api-v1-batch
- api-v1-business-data
- api-keys-manage
- webhooks-manage
- webhooks-deliver

Search (2 functions)

- universal-search
- search

Extraction (59 functions)

- bat-simple-extract, bat-extract, bat-queue-worker, bat-multisignal-postprocess, bat-url-discovery, bat-year-crawler
- extract-cars-and-bids-core, extract-cars-and-bids-comments
- extract-craigslist, extract-ebay-motors, extract-ebay-parts
- extract-facebook-marketplace, fb-marketplace-bot-scraper, fb-marketplace-orchestrator, fb-marketplace-sweep
- extract-hagerty-listing, hagerty-bid-tracker, hagerty-email-parser
- extract-premium-auction, extract-rmsothebys, extract-gooding, extract-bonhams, extract-historics-uk, extract-barrett-jackson, extract-gaa-classics
- extract-blocket, extract-leboncoin, extract-collecting-cars, extract-collecting-cars-simple
- extract-victorylap-listing, extract-bh-auction

- import-pcarmarket-listing, import-classic-auction, import-classiccars-listing, import-fb-marketplace
- extract-vehicle-data-ai, extract-vehicle-data-ollama, intelligent-scraper, unified-scraper-orchestrator
- extract-wayback-listing, extract-wayback-index, wayback-indexer
- [+ 18 more specialized extractors]

AI & Analysis (30+ functions)

- compute-vehicle-valuation, calculate-vehicle-scores, score-vehicle-condition
- analyze-auction-comments, analyze-comments-fast, analyze-batch-contextual
- identify-vehicle-from-image, validate-vehicle-image, analyze-image
- extract-vehicle-description, analyze-vehicle-description
- decode-vin-and-update, nlq-sql
- market-intelligence-agent, bid-curve-analysis, auction-trends-stats
- [+ 15 more]

Orchestration (15+ functions)

- ralph-wiggum-rlm-extraction-coordinator
- agent-orchestrator, autonomous-extraction-agent
- crawler-scheduler, continuous-queue-processor
- pipeline-orchestrator (GitHub Action)
- [+ 9 more]

Payments (8 functions)

- create-checkout, create-api-access-checkout, create-vehicle-transaction-checkout
- stripe-webhook, create-setup-session, setup-payment-method
- place-market-order, place-bid-with-deposit

Notifications (12+ functions)

- send-transaction-sms, sms-reminder-scheduler, sms-review
- telegram-intake, telegram-webhook, telegram-restoration-bot
- create-notification, concierge-notify
- [+ 4 more]

Platform (10+ functions)

- db-stats, platform-status, check-scraper-health, system-health-monitor
- queue-status, process-import-queue, process-content-extraction
- [+ 3 more]

---

29. Appendix C: Sample API Responses

Vehicle Lookup Response

GET /v1/vehicles/JT2AE86C9P0123456


```
{
  "id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "vin": "JT2AE86C9P0123456",
  "year": 1993,
  "year_confidence": 10,
  "year_source": "vin_decode",
  "make": "Toyota",
  "make_confidence": 10,
  "make_source": "vin_decode",
  "model": "MR2",
  "model_confidence": 10,
  "model_source": "vin_decode",
  "trim": "Turbo",
  "trim_confidence": 7,
  "trim_source": "bring_a_trailer",
  "body_style": "Coupe",
  "engine": "3S-GTE 2.0L Turbo I4",
  "transmission": "Manual",
  "drivetrain": "RWD",
  "color": "Super Red",
  "color_confidence": 8,
  "color_source": "bring_a_trailer",
  "mileage": 78420,
  "mileage_confidence": 7,
  "mileage_source": "bring_a_trailer",
  "mileage_observed_at": "2025-11-15T00:00:00Z",
  "images": {
    "count": 42,
    "primary": "https://storage.nuke.dev/vehicles/a1b2.../
        primary.jpg",
    "categories": {
      "exterior": 18,
      "interior": 10,
      "engine": 6,
      "undercarriage": 4,
      "detail": 4
    }
  },
  "observations_count": 23,
  "sources": ["bring_a_trailer", "facebook_marketplace",
        "rennlist_forum"],
  "first_observed": "2024-03-12T08:15:00Z",
  "last_observed": "2026-01-28T14:32:00Z",
  "valuation": {
```

```
    "estimated_value": 32500,
    "confidence": 7.8,
    "range": [27000, 38000],
    "based_on_comparables": 15,
    "trend_12mo": "+8.2%",
    "price_tier": "mainstream",
    "deal_score": null,
    "heat_score": 42,
    "heat_label": "warm",
    "computed_at": "2026-02-10T06:00:00Z"
  },
  "latest_listing": {
    "platform": "bring_a_trailer",
    "status": "sold",
    "final_price": 31500,
    "bid_count": 28,
    "sold_at": "2025-11-15T22:00:00Z",
    "url": "https://bringatrailer.com/listing/1993-toyota-mr2-
        turbo-42/"
  },
  "listing_count": 3,
  "auction_count": 1,
  "created_at": "2024-03-12T08:15:00Z",
  "updated_at": "2026-02-10T06:00:00Z"
}
```

Observation History Response

GET /v1/vehicles/JT2AE86C9P0123456/history?kind=sale_result,listing


```
{
  "vehicle_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "vin": "JT2AE86C9P0123456",
  "observations": [
    {
      "id": "obs_001",
      "kind": "sale_result",
      "source": "bring_a_trailer",
      "confidence": 0.92,
      "observed_at": "2025-11-15T22:00:00Z",
      "data": {
        "final_price": 31500,
        "bid_count": 28,
        "reserve_met": true,
        "auction_url": "https://bringatrailer.com/listing/..."
      },
      "source_url": "https://bringatrailer.com/listing/1993-toyota-
        mr2-turbo-42/"
    },
    {
      "id": "obs_002",
      "kind": "listing",
      "source": "facebook_marketplace",
      "confidence": 0.65,
      "observed_at": "2025-08-03T10:30:00Z",
      "data": {
        "asking_price": 35000,
        "listing_status": "expired",
        "days_listed": 34,
        "location": "Portland, OR"
      },
      "source_url": null
    },
    {
      "id": "obs_003",
      "kind": "listing",
      "source": "facebook_marketplace",
      "confidence": 0.62,
      "observed_at": "2025-03-18T15:45:00Z",
      "data": {
        "asking_price": 38500,
        "listing_status": "expired",
        "days_listed": 62,
        "location": "Portland, OR"
```

```
      },
      "source_url": null
    }
  ],
  "pagination": {
    "total": 3,
    "cursor": null,
    "has_more": false
  }
}
```

Market Comps Response

```
GET /v1/market/comps?
        make=Toyota&model=MR2&year_min=1991&year_max=1995&trim=Turbo
```


```
{
  "query": {
    "make": "Toyota",
    "model": "MR2",
    "year_range": [1991, 1995],
    "trim": "Turbo"
  },
  "summary": {
    "median_price": 34750,
    "mean_price": 36200,
    "min_price": 18500,
    "max_price": 67000,
    "total_transactions": 47,
    "time_range": "2024-02-10 to 2026-02-10",
    "trend_12mo": "+8.2%",
    "trend_confidence": 0.78
  },
  "transactions": [
    {
      "vehicle_id": "v_abc123",
      "year": 1993,
      "trim": "Turbo",
      "color": "Super Red",
      "mileage": 78420,
      "condition_notes": "Well-maintained, original paint, timing
        belt service at 72K",
      "sale_price": 31500,
      "platform": "bring_a_trailer",
      "sold_at": "2025-11-15",
      "bid_count": 28,
      "confidence": 0.92,
      "url": "https://bringatrailer.com/listing/..."
    }
  ],
  "pagination": {
    "total": 47,
    "cursor": "eyJwYWdlIjoxfQ==",
    "has_more": true
  }
}
```

30. Conclusion

Nuke has built the hard part: a continuously-running extraction pipeline spanning 25+ platforms, an observation-based data architecture with source provenance on every field, a confidence scoring framework, and an 800K+ vehicle dataset with 30M images.

What remains is packaging: exposing the strongest data through clean API endpoints, building developer experience, and going to market with a clear message.

The message is not “we have AI.” The message is “we went to the source.”

Every field in a Nuke API response tells you where it came from and how much to trust it. No other automotive data provider does this. In a market moving toward algorithmic decision-making — where lenders, insurers, and platforms need auditable, traceable, real-time data — this is the product.

The 90-day plan is straightforward: 1. **Month 1**: Ship the API surface (new endpoints over existing data) 2. **Month 2**: Ship the developer experience (docs, SDK, playground, billing) 3. **Month 3**: Ship to customers (design partners, case studies, iteration)

The data is flowing. The architecture is sound. The moat is deepening every day as more observations are recorded. Now it’s time to let the world plug in.

End of Report

Prepared by: Nuke Internal **Classification:** Confidential **Next Review:** March 2026